MiniCPM-V

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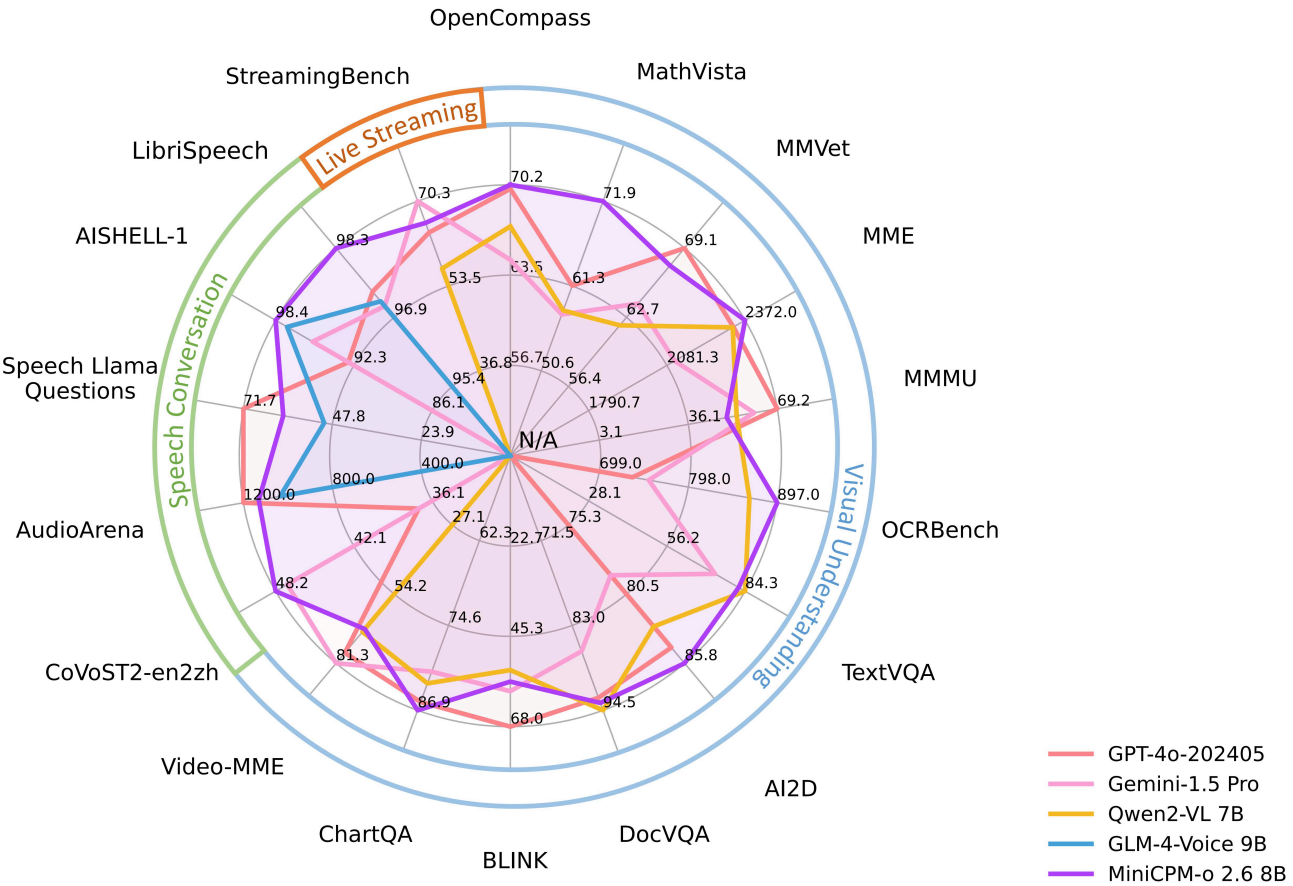
> **Demonstration environment**

Development board: Jetson Orin series mainboard

1. Model size

Model	Size
minicpm-v:8b	5.5GB

2. Performance



3. Using MiniCPM-V

3.1 Run MiniCPM-V

If the model is not present on the system, it pulls the minicpm-v:8b model automatically and runs it:

```
ollama run minicpm-v:8b
```

3.2 Start a conversation

```
Tell me a mathematician story
```

The time it takes to answer depends on the hardware configuration, so be patient!

```
>>> Tell me a mathematician story
One famous mathematician, Paul Erdős, was known for his unique approach to
collaboration and problem-solving. He believed that mathematics should be
approached with passion and creativity, rather than as just a set of rules
and formulas.
Erdős would often travel the world in search of new collaborators and
unsolved problems, working on them together in whatever spare time he had
available. His ability to connect with other mathematicians from all over
the globe was unparalleled, and many people credit him for helping to
advance mathematics as a collaborative enterprise rather than just an
academic pursuit.
Perhaps one of his most famous collaborations occurred when Erdős met
Andrew Odlyzko at a conference in Moscow during the Cold War. Despite
their political differences, they worked together on some complex
mathematical problems, leading to several important discoveries and
breakthroughs. Their friendship endured for decades, even surviving
through periods of intense international tension.
Overall, Paul Erdős was known as one of the greatest mathematicians of his
time, but it was his passion for collaboration that truly set him apart
from others in the field.
```

3.3 Vision function



```
What do you see in this picture? :./test_pic.png
# Using ": + the absolute path of the image" in the conversation lets the model use its
vision function and parse the information in the image
```

```
:~$ ollama run minicpm-v:8b
>>> What do you see in this picture? :./test_pic.png
Added image './test_pic.png'
This image shows two individuals on a rooftop with buildings and sky
visible. One person is sitting down, looking up at the sky while wearing a
white sleeveless top and shorts, holding what appears to be an empty
bottle of water. Another individual can be seen bending over something
near his feet in the background. The setting seems peaceful and relaxed
under bright sunlight.

>>> Send a message (/? for help)
```

3.4 End the conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

References

> **Ollama**

Official website: https://ollama.com/

GitHub: https://github.com/ollama/ollama

> **MiniCPM-V**

GitHub: https://github.com/OpenBMB/MiniCPM-o

Corresponding Ollama model: https://ollama.com/library/minicpm-v